

# Appendix: Choosing Better NLDL Layouts by Evaluating the Model in the High-dimensional Data Space

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# 1 Methods and hyper-parameters used to generate layouts

Table 1 contains the list of methods and hyper-parameters used for each of the layouts shown in the paper.

| Figure | NLDR<br>method | Hyper-parameter(s)                                               |
|--------|----------------|------------------------------------------------------------------|
| 1a     | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| 1b     | UMAP           | n_neighbors = 5, min_dist = 0.8                                  |
| 1c     | UMAP           | n_neighbors = 5, min_dist = 0.01                                 |
| 1d     | tSNE           | perplexity = 5                                                   |
| 1e     | tSNE           | perplexity = 30                                                  |
| 1f     | PHATE          | knn = 5                                                          |
| 1g     | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |
| 1h     | PaCMAP         | n_neighbors = 30, init = random, MN_ratio = 0.9,<br>FP_ratio = 5 |
| 2      | tSNE           | perplexity = 47                                                  |
| 4a     | tSNE           | perplexity = 47                                                  |
| 5b     | tSNE           | perplexity = 47                                                  |
| 6      | tSNE           | perplexity = 47                                                  |
| 8a     | tSNE           | perplexity = 47                                                  |
| 8b     | tSNE           | perplexity = 62                                                  |
| 8c     | UMAP           | n_neighbors = 15, min_dist = 0.1                                 |
| 8d     | PHATE          | knn = 5                                                          |

(continued)

| Figure | NLDR<br>method | Hyper-parameter(s)                                               |
|--------|----------------|------------------------------------------------------------------|
| 8e     | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |
| 8f     | PaCMAP         | n_neighbors = 10, init = random, MN_ratio = 0.5,<br>FP_ratio = 2 |
| 10a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| 10b    | UMAP           | n_neighbors = 5, min_dist = 0.8                                  |
| 10c    | UMAP           | n_neighbors = 5, min_dist = 0.01                                 |
| 10d    | tSNE           | perplexity = 5                                                   |
| 10e    | tSNE           | perplexity = 30                                                  |
| 10f    | PHATE          | knn = 5                                                          |
| 10g    | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |
| 10h    | PaCMAP         | n_neighbors = 30, init = random, MN_ratio = 0.9,<br>FP_ratio = 5 |
| 11a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| 11e    | tSNE           | perplexity = 30                                                  |
| 12a    | tSNE           | perplexity = 30                                                  |
| 12b    | tSNE           | perplexity = 89                                                  |
| 12c    | UMAP           | n_neighbors = 15, min_dist = 0.1                                 |
| 12d    | PHATE          | knn = 5                                                          |
| 12e    | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |
| 12f    | PaCMAP         | n_neighbors = 10, init = random, MN_ratio = 0.5,<br>FP_ratio = 2 |

(continued)

| Figure | NLDR<br>method | Hyper-parameter(s)                                               |
|--------|----------------|------------------------------------------------------------------|
| 13a    | tSNE           | perplexity = 30                                                  |
| 14a    | tSNE           | perplexity = 30                                                  |
| A4a    | tSNE           | perplexity = 71                                                  |
| A4b    | UMAP           | n_neighbors = 15, min_dist = 0.1                                 |
| A4c    | PaCMAP         | n_neighbors = 10, init = random, MN_ratio = 0.5,<br>FP_ratio = 2 |
| A5     | tSNE           | perplexity = 52                                                  |
| A6a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| A6b    | tSNE           | perplexity = 30                                                  |
| A7a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| A7b    | tSNE           | perplexity = 30                                                  |
| A8a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| A8b    | tSNE           | perplexity = 30                                                  |
| A9a    | UMAP           | n_neighbors = 30, min_dist = 0.3                                 |
| A9b    | UMAP           | n_neighbors = 5, min_dist = 0.8                                  |
| A9c    | UMAP           | n_neighbors = 5, min_dist = 0.01                                 |
| A9d    | tSNE           | perplexity = 5                                                   |
| A9e    | tSNE           | perplexity = 30                                                  |
| A9f    | PHATE          | knn = 5                                                          |
| A9g    | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |



(continued)

| Figure | NLDR<br>method | Hyper-parameter(s)                                               |
|--------|----------------|------------------------------------------------------------------|
| A9h    | PaCMAP         | n_neighbors = 30, init = random, MN_ratio = 0.9,<br>FP_ratio = 5 |
| A10a   | tSNE           | perplexity = 30                                                  |
| A10b   | tSNE           | perplexity = 89                                                  |
| A10c   | UMAP           | n_neighbors = 15, min_dist = 0.1                                 |
| A10d   | PHATE          | knn = 5                                                          |
| A10e   | TriMAP         | n_inliers = 12, n_outliers = 4, n_random = 3                     |
| A10f   | PaCMAP         | n_neighbors = 10, init = random, MN_ratio = 0.5,<br>FP_ratio = 2 |

Table 1: NLDR methods and hyper-parameters used for each Figure in the main paper.

## 2 Videos links

Animations of the  $p$ - $D$  tours that produced specific projections shown in some figures in the main paper are available on YouTube at the links given in Table 2.

| Figure | URL                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| 4      | <a href="https://youtu.be/yHKTHK4UBiU">youtu.be/yHKTHK4UBiU</a>                                                                   |
| 5      | <a href="https://youtu.be/Fukiminr090">youtu.be/Fukiminr090</a>                                                                   |
| 11     | <a href="https://youtu.be/3VfK3M2gnZM">youtu.be/3VfK3M2gnZM</a> , <a href="https://youtu.be/Es84bwQcndU">youtu.be/Es84bwQcndU</a> |
| 13     | <a href="https://youtu.be/sUcGd57Swdg">youtu.be/sUcGd57Swdg</a> , <a href="https://youtu.be/QiklCjELUxo">youtu.be/QiklCjELUxo</a> |

Table 2: Videos of the langevitour animations and the linked plots.

### 3 Notation

| Notation                 | Description                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------|
| $n, p, k$                | number of observations, variables, embedding dimension, respectively                        |
| $\mathbf{X}, \mathbf{x}$ | $p$ -dimensional data (population, sample)                                                  |
| $\mathbf{y}$             | $k$ -dimensional layout                                                                     |
| $P$                      | orthonormal basis, generating a $d$ -dimensional linear projection of $p$ -dimensional data |
| $T$                      | true model                                                                                  |
| $g$                      | functional mapping from $p$ -D to $k$ -D, especially as prescribed by NLDR                  |
| $\theta$                 | (Hyper-) parameters for NLDR method                                                         |
| $r$                      | ranges of the embedding components                                                          |
| $C^{(j)}$                | $j$ -dimensional bin centers                                                                |
| $(b_1, b_2)$             | number of bins in each direction                                                            |
| $(a_1, a_2)$             | binwidths, distance between centroids in each direction                                     |
| $(s_1, s_2)$             | starting coordinates of the hexagonal grid                                                  |
| $q$                      | buffer to ensure hexgrid covers data, proportion of data range, 0-1                         |
| $m$                      | number of non-empty bins                                                                    |
| $b$                      | number of hexagons in the grid                                                              |
| $h$                      | hexagonal id                                                                                |
| $H_h$                    | set of observations in bin $h$                                                              |
| $l$                      | side length                                                                                 |
| $A$                      | area                                                                                        |
| $n_h$                    | number of points in hexagon $h$ (bin count)                                                 |
| $w_h$                    | standardized number of points in hexagon $h$ (standardized bin counts)                      |

Table 3: Summary of notation for describing new methodology.

### 4 Scripts

| Folder | Script                 | Description                                |
|--------|------------------------|--------------------------------------------|
| script | additional_functions.R | Helper functions to render the main paper. |

(continued)

| Folder        | Script              | Description                                                                                                        |
|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------|
| script        | evaluation.py       | Python script implementing additional evaluation metrics such as RTA and GS.                                       |
| script        | nldr_code.R         | Wrapper functions for running multiple NLDR methods (UMAP, tSNE, PHATE, PaCMAP, TriMAP) with different parameters. |
| two_nonlinear | 01_gen_data.R       | Generates the 2NC7 dataset.                                                                                        |
| two_nonlinear | 02_gen_true_model.R | Creates the true structure of 2NC7 data.                                                                           |

(continued)

| Folder        | Script                          | Description                                                                                |
|---------------|---------------------------------|--------------------------------------------------------------------------------------------|
| two_nonlinear | 03_gen_embeddings.R             | Computes multiple NLDR embeddings for the 2NC7 data.                                       |
| two_nonlinear | 04_gen_mse_for_diff_methods.R   | Computes HBE with varying bin widths ( $a_1$ ) for all NLDR embeddings.                    |
| two_nonlinear | 05_gen_rm_lwd_mse.R             | Computes HBE with varying low density bin cutoff for all three binwidth ( $a_1$ ) choices. |
| two_nonlinear | 06_gen_model_with_tSNE.R        | Fits the model for the layout a.                                                           |
| two_nonlinear | 07_example_evaluation_metrics.R | Calculates evaluation metrics for all NLDR layouts.                                        |

(continued)

| Folder            | Script                                   | Description                                           |
|-------------------|------------------------------------------|-------------------------------------------------------|
| two_nonlinear     | 08_gen_model_with_PHATE.R                | Fits the model<br>for the layout c.                   |
| five_gau_clusters | 01_five_gaussian_clus-<br>ter_data_emb.R | Generates data<br>and multiple<br>NLDR<br>embeddings. |
| five_gau_clusters | 02_gen_model_with_tSNE.R                 | Fits the model<br>for the layout a.                   |
| five_gau_clusters | 03_gen_model_with_UMAP.R                 | Fits the model<br>for the layout b.                   |
| five_gau_clusters | 04_gen_model_with_PaCMAP.R               | Fits the model<br>for the layout c.                   |
| c_shaped_dens_str | 01_gen_data.R                            | Generates the<br>2-D curved<br>sheet dataset.         |
| c_shaped_dens_str | 02_gen_embeddings_uni_dens.R             | Generates<br>multiple NLDR<br>embeddings.             |
| c_shaped_dens_str | 03_gen_model_with_tSNE.R                 | Fits the model<br>for the tSNE<br>layout.             |

(continued)

| Folder | Script                   | Description                                                              |
|--------|--------------------------|--------------------------------------------------------------------------|
| pbmc3k | 01_obtain_pca_author.R   | Obtains author's PCA results.                                            |
| pbmc3k | 02_obtain_umap_authors.R | Obtains author's UMAP embeddings.                                        |
| pbmc3k | 03_gen_umap_diff_param.R | Generates multiple UMAP embeddings with different hyperparameter values. |
| pbmc3k | 04_gen_tsne_diff_param.R | Generates multiple tSNE embeddings with different hyperparameter values. |
| pbmc3k | 05_gen_phate.R           | Generates a PHATE embeddings with default hyperparameters.               |

(continued)

| Folder | Script                        | Description                                                             |
|--------|-------------------------------|-------------------------------------------------------------------------|
| pbmc3k | 06_gen_trimap.R               | Generates a TriMAP embeddings with default hyper-parameters.            |
| pbmc3k | 07_gen_pacmap.R               | Generates a PaCMAP embeddings with default hyper-parameters.            |
| pbmc3k | 08_gen_mse_for_diff_methods.R | Computes HBE with varying bin widths ( $a_1$ ) for all NLDR embeddings. |
| pbmc3k | 09_gen_scDEED.R               | Generates UMAP embeddings from scDEED results.                          |

(continued)

| Folder | Script                         | Description                                                                     |
|--------|--------------------------------|---------------------------------------------------------------------------------|
| pbmc3k | 10_pre_process_for_embedding.R | Generates<br>PBMC3k data<br>used for<br>scDEED results.                         |
| pbmc3k | 11_gen_mse_for_diff_tsne_scD.R | Computes HBE<br>with varying bin<br>widths ( $a_1$ ) for<br>tSNE<br>embeddings. |
| pbmc3k | 12_gen_mse_for_diff_umap_scD.R | Computes HBE<br>with varying bin<br>widths ( $a_1$ ) for<br>UMAP<br>embeddings. |
| pbmc3k | 13_gen_model_with_UMAP.R       | Fits the model<br>for the layout a.                                             |
| pbmc3k | 14_gen_model_with_tSNE.R       | Fits the model<br>for the layout e.                                             |
| pbmc3k | 15_gen_model_with_UMAP_scD.R   | Fits the model<br>for the layout a.                                             |
| pbmc3k | 16_gen_model_with_tSNE_scD.R   | Fits the model<br>for the layout b.                                             |



(continued)

| Folder | Script                        | Description                                                             |
|--------|-------------------------------|-------------------------------------------------------------------------|
| pbmc3k | 17_evaluation_metrics.R       | Calculates evaluation metrics for all NLDR layouts.                     |
| pbmc3k | 18_evaluation_metrics_scD.R   | Calculates evaluation metrics for all NLDR layouts.                     |
| mnist  | 01_data_preprocessing.R       | Computes first 10 principal components and save data.                   |
| mnist  | 02_gen_diff_embeddings.R      | Generates multiple NLDR embeddings.                                     |
| mnist  | 03_gen_mse_for_diff_methods.R | Computes HBE with varying bin widths ( $a_1$ ) for all NLDR embeddings. |
| mnist  | 04_gen_model_with_tSNE.R      | Fits the model for the layout a.                                        |

(continued)

| Folder | Script                   | Description                                                  |
|--------|--------------------------|--------------------------------------------------------------|
| mnist  | 05_evaluation_metrics.R  | Calculates<br>evaluation<br>metrics for all<br>NLDR layouts. |
| mnist  | 06_link_brush_layout_e.R | Creates<br>interactive<br>linked brushing<br>with layout e.  |

Table 4: R and Python script files used to generate outputs in the main paper.

## 5 Generating the 2NC7 data

This data is constructed by simulating two clusters, each consisting of 1000 observations. The C-shaped cluster is generated from  $\theta \sim U(-3\pi/2, 0)$ ,  $X_1 = \sin(\theta)$ ,  $X_2 \sim U(0, 2)$  (adding thickness to the C),  $X_3 = \text{sign}(\theta) \times (\cos(\theta) - 1)$ ,  $X_4 = \cos(\theta)$ . Observations lie on a 2- $D$  manifold in 7- $D$ . The other cluster is from  $X_1 \sim U(0, 2)$ ,  $X_2 \sim U(0, 3)$ ,  $X_3 = -(X_1^3 + X_2)$ , and  $X_4 \sim U(0, 2)$ . It is also curved, but observations lie on a 3- $D$  manifold in 7- $D$ . Three more variables,  $X_5, X_6, X_7$ , that are small amounts of pure noise are added. We would consider  $T = (X_1, X_2, X_3, X_4)$  to be the geometric structure (true model) that we hope to capture.

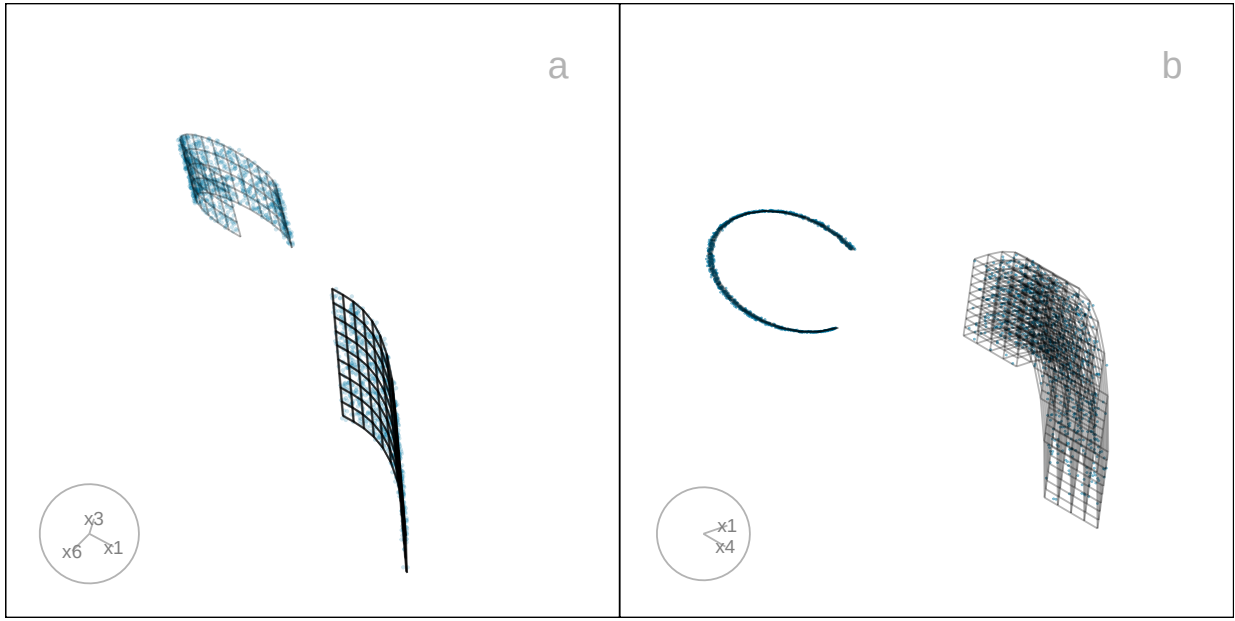


Figure 1: Two projections of the  $p$ -D true model overlaying the data are shown in a, b. Video of the langevitour animations is available at <https://youtu.be/35TrnYJsUUI>.

## 6 Computing hexagon grid configurations

Given range of embedding component,  $r_2$ , number of bins along the x-axis,  $b_1$ , and buffer proportion,  $q$ , hexagonal starting point coordinates,  $s_1 = -q$ , and  $s_2 = -qr_2$ . The purpose is to find width of the hexagon,  $a_1$  and number of bins along the y-axis,  $b_2$ .

Geometric arguments give rise to the following constraints.

$\min a_1$  s.t.

$$s_1 - \frac{a_1}{2} < 0, \quad (1)$$

$$s_1 + (b_1 - 1) \times a_1 \geq 1, \quad (2)$$

$$s_2 - \frac{a_2}{2} < 0, \quad (3)$$

$$s_2 + (b_2 - 1) \times a_2 \geq r_2. \quad (4)$$

Since  $a_1$  and  $a_2$  are distances,

$$a_1, a_2 > 0.$$

Also,  $(s_1, s_2) \in (-0.1, -0.05)$  as these are multiplicative offsets in the negative direction.

Equation 1 can be rearranged as,

$$a_1 > 2s_1$$

which given  $s_1 < 0$  and  $a_1 > 0$  will *always* be true. The same logic follows for Equation 3

and substituting  $a_2 = \sqrt{3}a_1/2$ , and  $s_2 = -qr_2$  to Equation 3 can be written as,

$$a_1 > -\frac{4}{\sqrt{3}}qr_2$$

Also, substituting  $a_2 = \sqrt{3}a_1/2$ ,  $s_2 = -qr_2$  and rearranging Equation 4 gives:

$$a_1 \geq \frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)}. \quad (5)$$

Similarly, substituting  $s_1 = -q$  Equation 2 becomes,

$$a_1 \geq \frac{(1+q)}{(b_1-1)}. \quad (6)$$

This is a linear optimization problem. Therefore, the optimal solution must occur on a vertex. So, by setting Equation 5 equals to Equation 6 gives,

$$\frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)} = \frac{(1+q)}{(b_1 - 1)}.$$

After rearranging this,

$$b_2 = 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}}$$

and since  $b_2$  should be an integer,

$$b_2 = \left\lceil 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}} \right\rceil. \quad (7)$$

Furthermore, with known  $b_1$  and  $b_2$ , by considering Equation 2 or Equation 4 as the *binding* or *active constraint*, can compute  $a_1$ .

If Equation 2 is active, then,

$$\frac{(1+q)}{(b_1 - 1)} < \frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)}.$$

Rearranging this gives,

$$r_2 > \frac{\sqrt{3}(b_2 - 1)}{2(b_1 - 1)}.$$

Therefore, if this equality is true, then

$$a_1 = \frac{(1 + q)}{(b_1 - 1)},$$

otherwise,

$$a_1 = \frac{2r_2(1 + q)}{\sqrt{3}(b_2 - 1)}.$$

## 7 Binning the data

Points are assigned to the bin they fall into based on the nearest centroid. If a point is equidistant from multiple centroids, it is assigned to the centroid with the smallest bin ID.

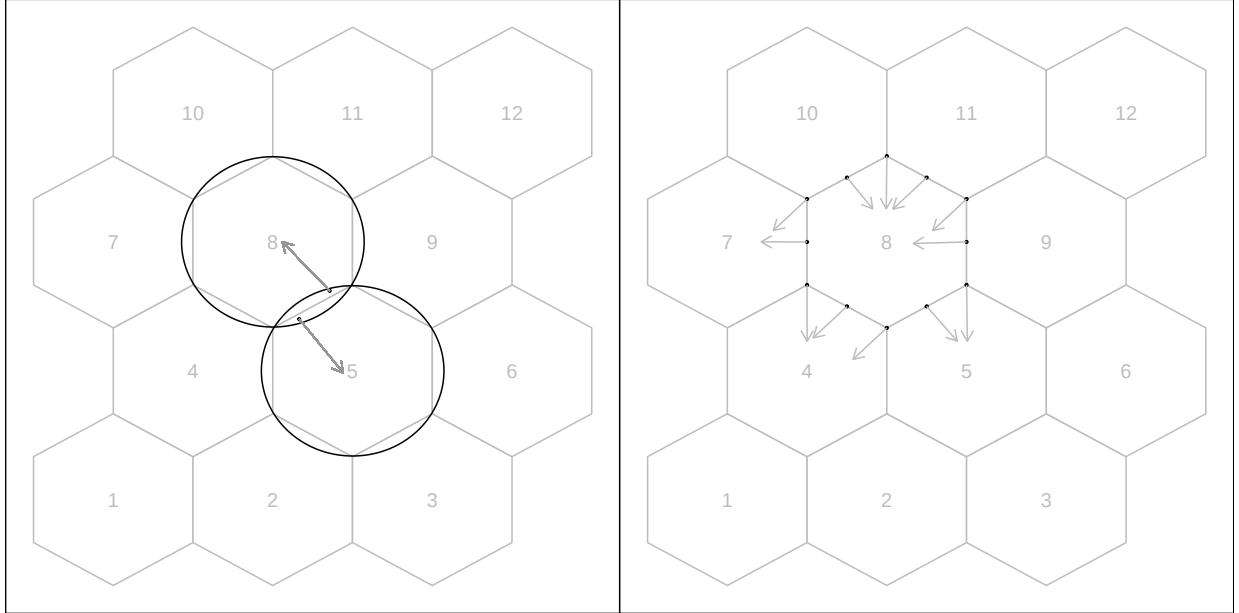


Figure 2: Binning the data. Points are assigned to the nearest centroid. If a point is equidistant from multiple centroids, assigned to the centroid with the smallest bin ID.

## 8 Area of a hexagon

The area of a hexagon is defined as  $A = 3\sqrt{3}l^2/2$ , where  $l$  is the side length of the hexagon.

$l$  can be computed using  $a_1$  and  $a_2$ .

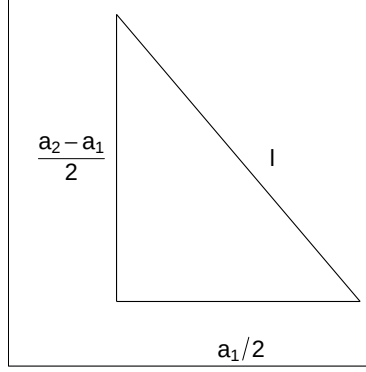


Figure 3: The components of the right triangle illustrating notation.

By applying the Pythagorean theorem, we obtain,

$$l^2 = \left(\frac{a_1}{2}\right)^2 + \left(\frac{a_2 - l}{2}\right)^2.$$

Next, rearranging the terms, we get,

$$l^2 - \left(\frac{a_2 - l}{2}\right)^2 = \left(\frac{a_1}{2}\right)^2,$$

$$\left[l - \left(\frac{a_2 - l}{2}\right)\right] \left[l + \left(\frac{a_2 - l}{2}\right)\right] = \left(\frac{a_1}{2}\right)^2,$$

$$3l^2 + 2a_2l - (a_1^2 + a_2^2) = 0.$$

Finally, by solving the quadratic equation, we compute,

$$l = \frac{-2a_2 \pm \sqrt{4a_2^2 - 24[-(a_1^2 + a_2^2)]}}{6},$$

$$l = \frac{-a_2 \pm \sqrt{a_2^2 - 6[-(a_1^2 + a_2^2)]}}{3},$$

where  $l > 0$ .

## 9 Curiosities about NLDR results discovered by examining the model in the data space

With the drawing of the model in the data, several interesting differences between NLDR methods can be observed.

### 9.1 Some methods appear to order points in the layout

The 2- $D$  model representations generated from some NLDR methods, especially PaCMAP, are unreasonably flat or like a pancake. A simple example of this can be seen with data simulated to contain five 4- $D$  Gaussian clusters. Each cluster is essentially a ball in 4- $D$ , so there is no 2- $D$  representation, rather the model in each cluster should resemble a crumpled sheet of paper that fills out 4- $D$ .

Figure 4 a1, b1, c1 show the 2- $D$  layouts for (a) tSNE, (b) UMAP, and (c) PaCMAP, respectively. The default hyper-parameters for each method are used. In each layout we can see an accurate representation where all five clusters are visible, although with varying degrees of separation.



The models are fitted to each these layouts. Figure 4 a2, b2, c2 show the fitted models in a projection of the 4- $D$  space, taken from a tour. These clusters are fully 4- $D$  in nature, so we would expect the model to be a *crumpled sheet* that stretches in all four dimensions. This is what is mostly observed for tSNE and UMAP. The curious detail is that the model for PaCMAP is closer to a *pancake* in shape in every cluster! This single projection only shows this in three of the five clusters but if we examine a different projection the other clusters exhibit the pancake also. While we don't know what exactly causes this, it is likely due to some ordering of points in the 2- $D$  PaCMAP layout that induces the flat model. One could imagine that if the method used principal components on all the data, that it might induce some ordering that would produce the flat model. If this were the reason, the pancaking would be the same in all clusters, but it is not: The pancake is visible in some clusters in some projections but in other clusters it is visible in different projections. It might be due to some ordering by nearest neighbors in a cluster. The PaCMAP documentation doesn't provide any helpful clues. That this happens, though, makes the PaCMAP layout inadequate for representing the high-dimensional data.

## 9.2 Sparseness creates a contracted 2- $D$ layout

Differences in density can arise by sampling at different rates in different subspaces of  $p$ - $D$ . For example, the data shown in Figure 5 all lies on a 2- $D$  curved sheet in 4- $D$ , but one end of the sheet is sampled densely and the other very sparsely. It was simulated to illustrate the effect of the density difference on layout generated by an NLDR, illustrated using the tSNE results, but it happens with all methods.

Figure 5 (a2, b2) shows a 2- $D$  layout for tSNE created using the default hyper-parameters. One would expect to see a rectangular shape if the curved sheet is flattened, but the layout

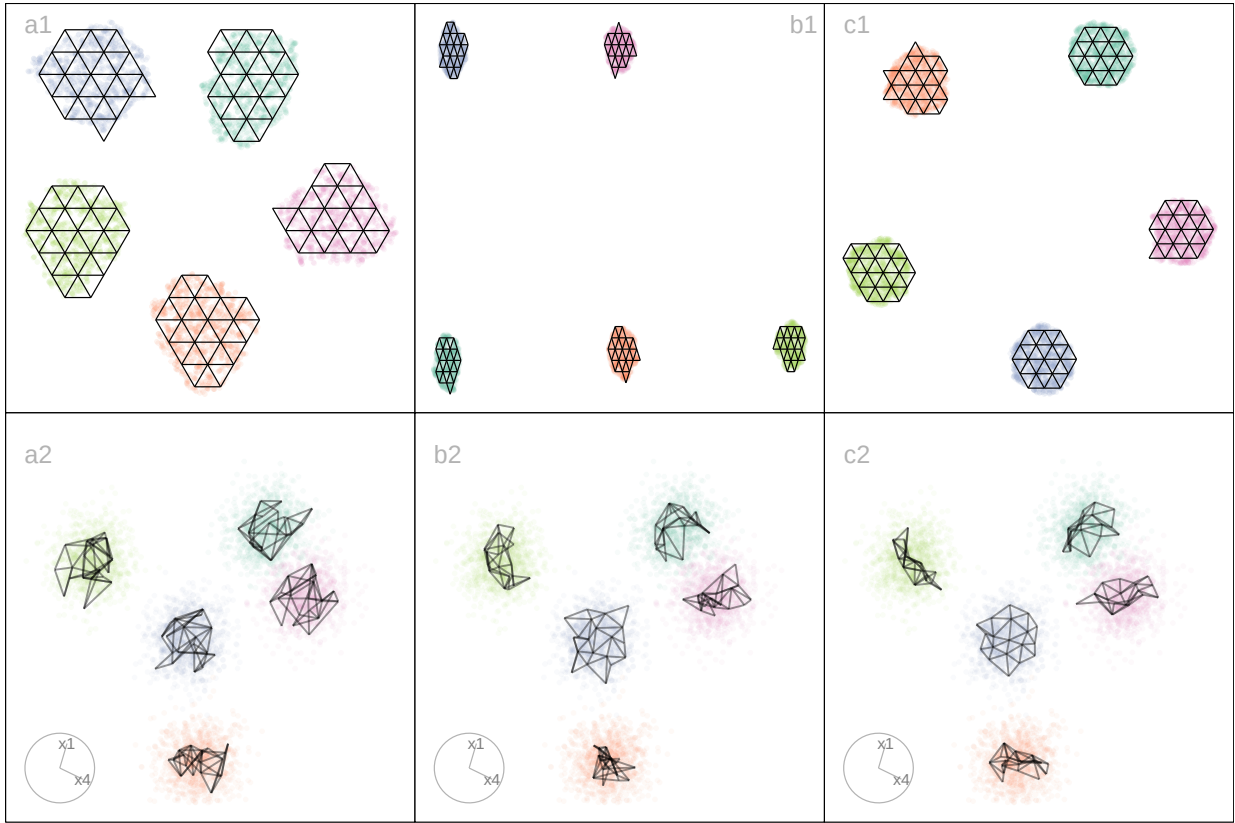


Figure 4: NLDR’s organize points in the 2- $D$  layout in different ways, possibly misleadingly, illustrated using three layouts: (a) tSNE, (b) UMAP, (c) PaCMAP. The data has five Gaussian clusters in 4- $D$ . The bottom row of plots shows a 2- $D$  projection from a tour on 4- $D$  revealing the differences generated by the layouts on the model fits. We would expect the model fit to be like that in (a2) where it is distinctly is separate for each cluster but like a hairball in each. This would indicate the distinct clusters, each being fully 4- $D$ . With (c2), the curiosity is that the model is a 2- $D$  pancake shape in 4- $D$ , indicating that there is some ordering of points done by PaCMAP, possibly along some principal component axes. Videos of the langevitour animations are available at <https://youtu.be/I-kxCwVfqiQ>, <https://youtu.be/gD1P01FUPyU>, and [https://youtu.be/MxJ\\_srOFQNk](https://youtu.be/MxJ_srOFQNk) respectively.

is triangular. The other two displays show the residuals as a dot density plot (a1, b1), and a 2- $D$  projection of the data and the model from 4- $D$  (a3, b3). Using linked brushing between the plots, we can highlight points in the tSNE layout, and examine where they fall in the original 4- $D$ . The darker (maroon) points indicate points that have been highlighted by linking. In row a, the points at the top of the triangle are highlighted, and we can see these correspond to higher residuals, and also to all points at the low density end of the curved sheet. In row b, points at the lower left side of the triangle are highlighted which

corresponds to smaller residuals and one corner of the sheet at the high density end of the curved sheet.

The tSNE behaviour is to squeeze the low density area of the data together into the layout. This is common in other NLDR methods also, which means analysts need to be aware that if their data is not sampled relatively uniformly, apparent closeness in the 2- $D$  may correspond to sparseness in  $p$ - $D$ .

## 10 PBMC3k: comparison with results of scDEED recommendations

As we were writing this paper [Xia et al. \(2023\)](#) appeared proposing a new method called scDEED helping to assess the validity of a 2- $D$  embedding. scDEED calculates a reliability score for each cell embedding based on the similarity between the cell’s 2- $D$  embedding neighbors and its neighbors prior to embedding. A low reliability score suggests a dubious embedding. It can help in the deciding on optimal hyper-parameters. Here we illustrate how our method compares with the results from scDEED.

Note that [Xia et al. \(2023\)](#) uses a different PBMC dataset than that used by [Chen et al. \(2024\)](#), shown by us in the main paper example, which is why this comparison is shown here and not in the main paper. Their data contains 31,021 cells including cell type labels, and the gene expression levels were in the unit of log-transformed UMI count per 10,000. They focused on three sequencing methods (inDrops, DropSeq, and SeqWell) and four common cell types Cytotoxic T cell, CD4+T cell, CD14+ Monocyte, and B cell. Pre-processing follows the process in [Xia et al. \(2023\)](#) again using the Human Peripheral Blood Mononuclear Cells (PBMC) data.

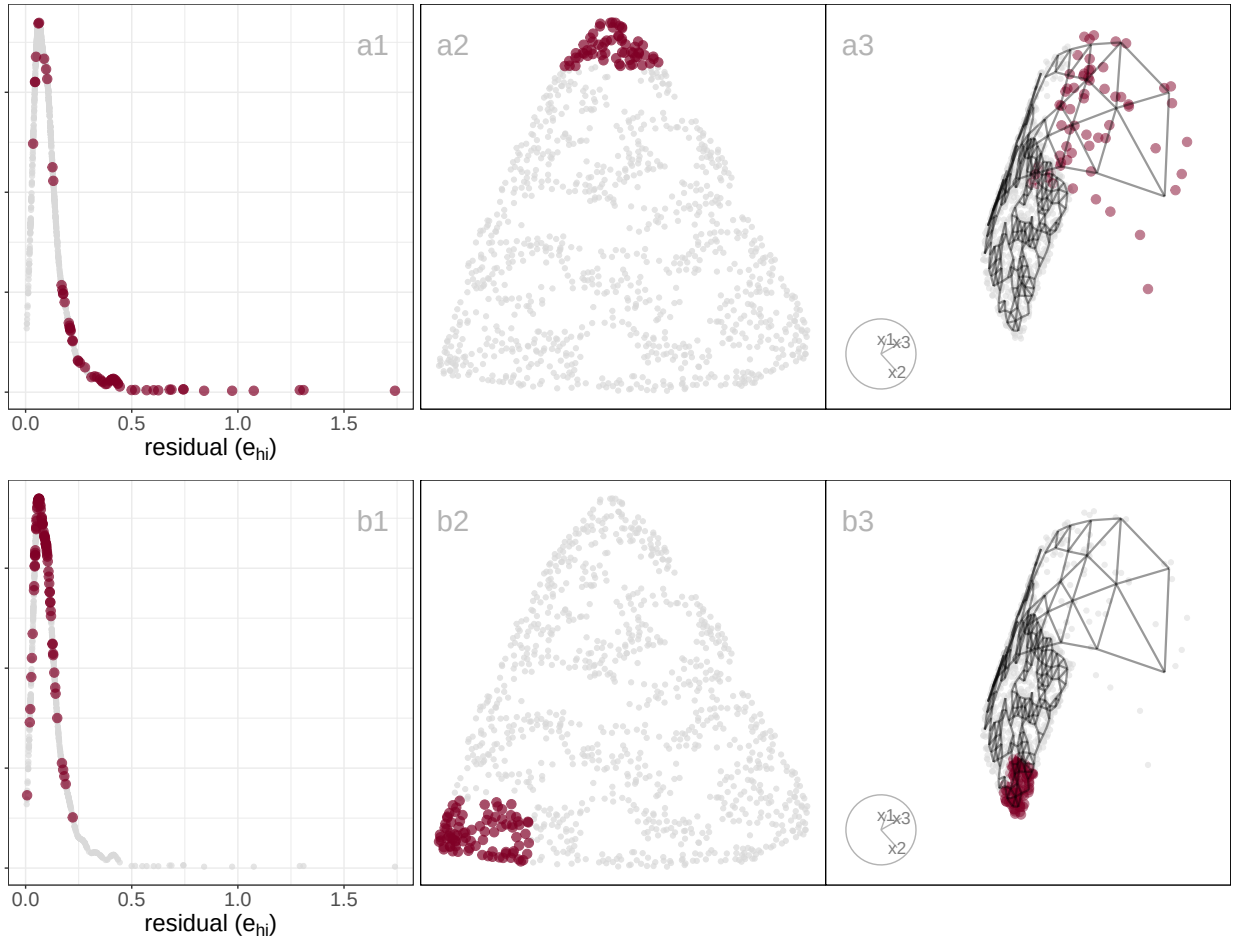


Figure 5: Exploring the effect of density on the NLDR layout using a 2- $D$  curved sheet in 4- $D$  with different density at each end. Three plots are linked: density plot of residuals (a1, b1), NLDR layout (a2, b2), projection of 4- $D$  model and data (a3, b3). The brown points indicate the selected set, which are different in each row. In (a2), the top part of the triangular shape is selected which corresponds to higher residuals (a1) and the sparse end of the structure (a3). In (b2) one of other corners is highlighted, which can be seen to correspond to low residuals (b1) and one side of the dense end of the data (b3). While the tSNE layout represents the dense end of the sheet correctly as two corners in the layout, it contracts the sparse end of the sheet into a single corner. Video of the langevitour animation is available at <https://youtu.be/-KsQH0rII2A>.

For illustration purposes, we only selected cells generated with inDrops ( $n = 5858$  cells).

Also, [Xia et al. \(2023\)](#) used first 9 principal components to generate the UMAP and tSNE with default hyper-parameters. The objective is to what scDEED suggests is the best layout with what HBE would choose. Layout a (Figure 6) is generated from the hyper-parameters suggested by [Chen et al. \(2024\)](#), and layout b (Figure 6) is with suggested hyper-parameters

by scDEED to be more accurate. The HBE vs binwidth ( $a_1$ ) plot (Figure 6) illustrates that our approach would suggest that scDEED is correct here, that layout b is more accurately reflecting the cluster structure in the PBMC data. This is also supported by examining the models in the data space as shown in Figure 7.

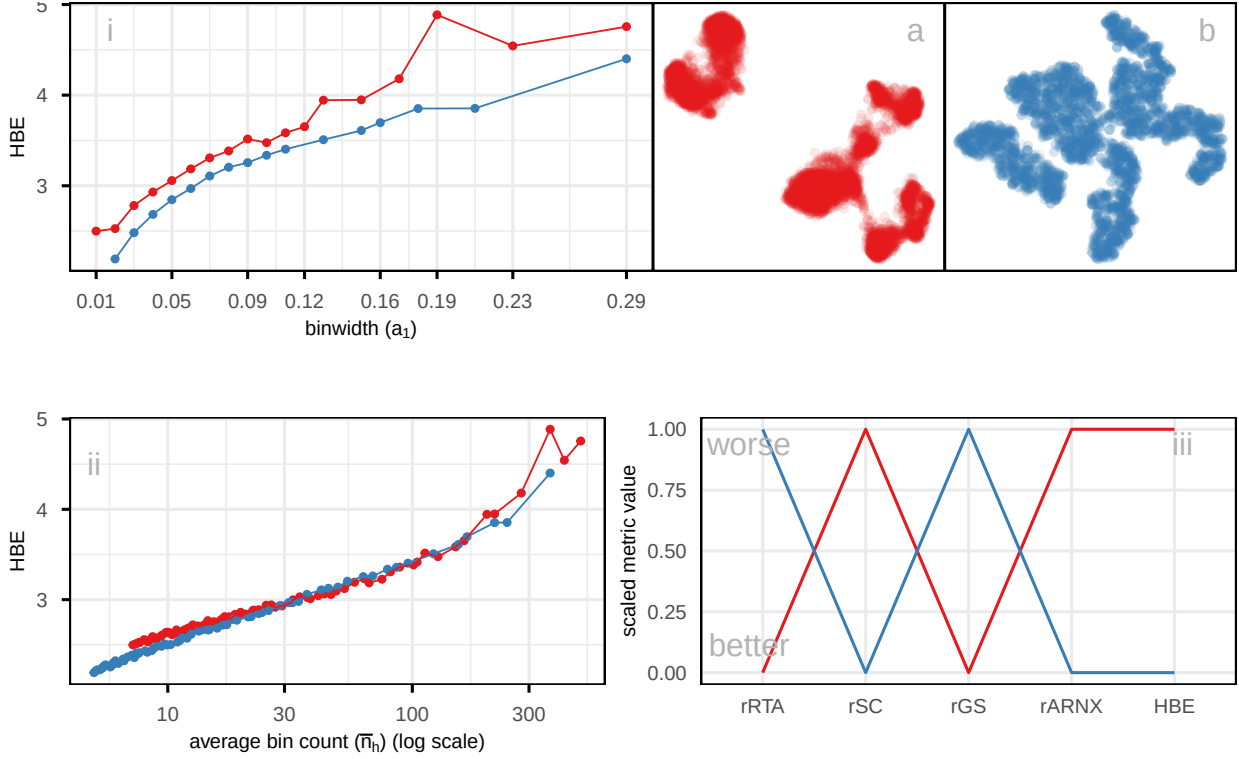


Figure 6: Comparing the published layout (a) with what would be suggested to be optimal by scDEED (b), using HBE for varying (i) binwidth ( $a_1$ ), and (ii) average bin count ( $\bar{n}_h$ ), on a subset of PBMC3k data. Color represents NLDR layouts. HBE would corroborate that the scDEED optimized layout is better than what was originally published. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences. Comparison of scaled evaluation metrics (iii) (rRTA, rSC, rGS, rARNX, and HBE using  $a_1 = 0.04$ ) for two NLDR layouts of the PBMC3k data the originally published layout (a) and the scDEED optimized layout (b). Each line represents a layout, with color matching the corresponding scatterplots. Most metrics (rSC, rARNX, and HBE) consistently indicate that the optimized layout (b) provides a better representation, while rRTA, and rGS slightly favor the published layout.

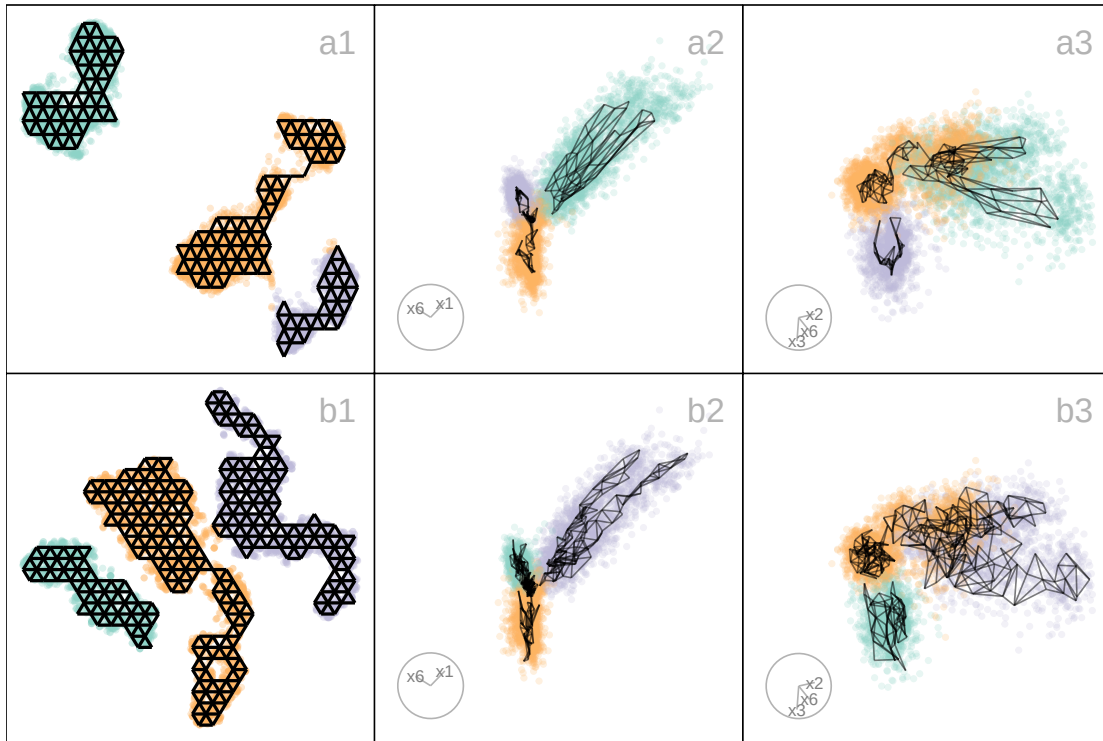


Figure 7: Compare the published 2- $D$  layout (Figure 6 a) made with UMAP and the 2- $D$  layout made with tSNE selected as optimal by scDEED (Figure 6 b) and also HBE (Figure 6). The two plots on the right show projections from a tour, with the models overlaid. The published layout a suggested three separated clusters with two of them are close, but this is not present in the data. While there may be three clusters they are not well-separated. The difference in model fit also indicates this: the published layout a does not capture the nonlinear structure of the clusters like the model generated from layout b. This supports the choice that layout b is the better representation of the data, because it shows close clusters. Videos of the langevitour animations are available at <https://youtu.be/ffiB4MGWyn8> and <https://youtu.be/e7XNL18co1c> respectively.

## 11 Compare HBE with existing evaluation metrics

Figure 8 and Figure 9 compare HBE with commonly used evaluation metrics such as rRTA, rARNX, rSC, and rGS across multiple NLDR layouts. These visual comparisons highlight that HBE behaves differently from these existing metrics due to the different settings involved.

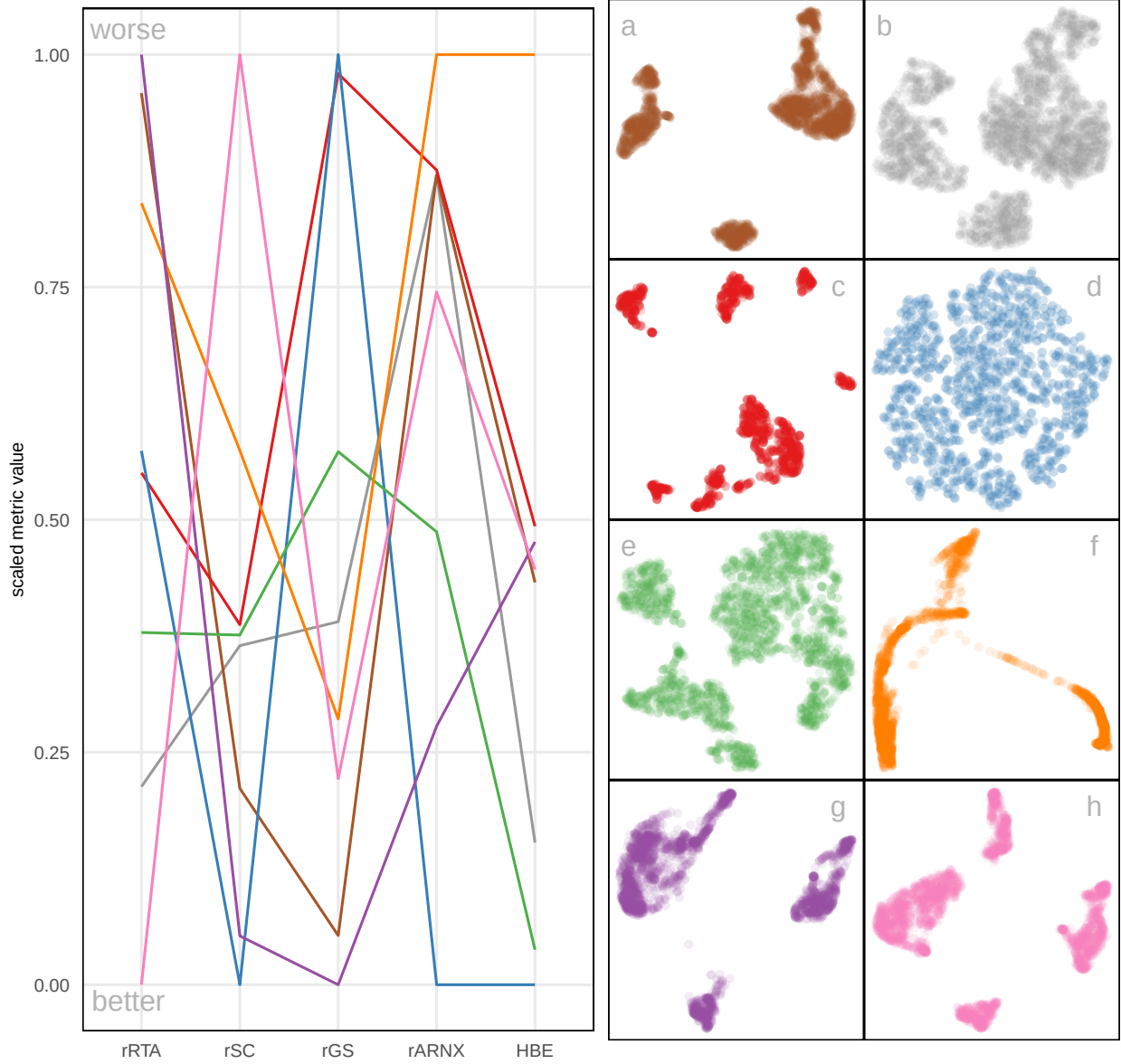


Figure 8: Comparison of scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE with  $a_1 = 0.06$ ) for the eight NLDR layouts computed on the PBMC3k data, shown as a parallel coordinate plot. The color of each line corresponds to a NLDR layout. All, except rGS and rARNX agree that layout e is best or very close to best. Layout d is best according to HBE and rARNX but considered to be much less optimal by rRTA, rSC and rGS. Layout f is considered poor by rARNX and HBE. Layout a is considered close to the best by rGS and rSC.



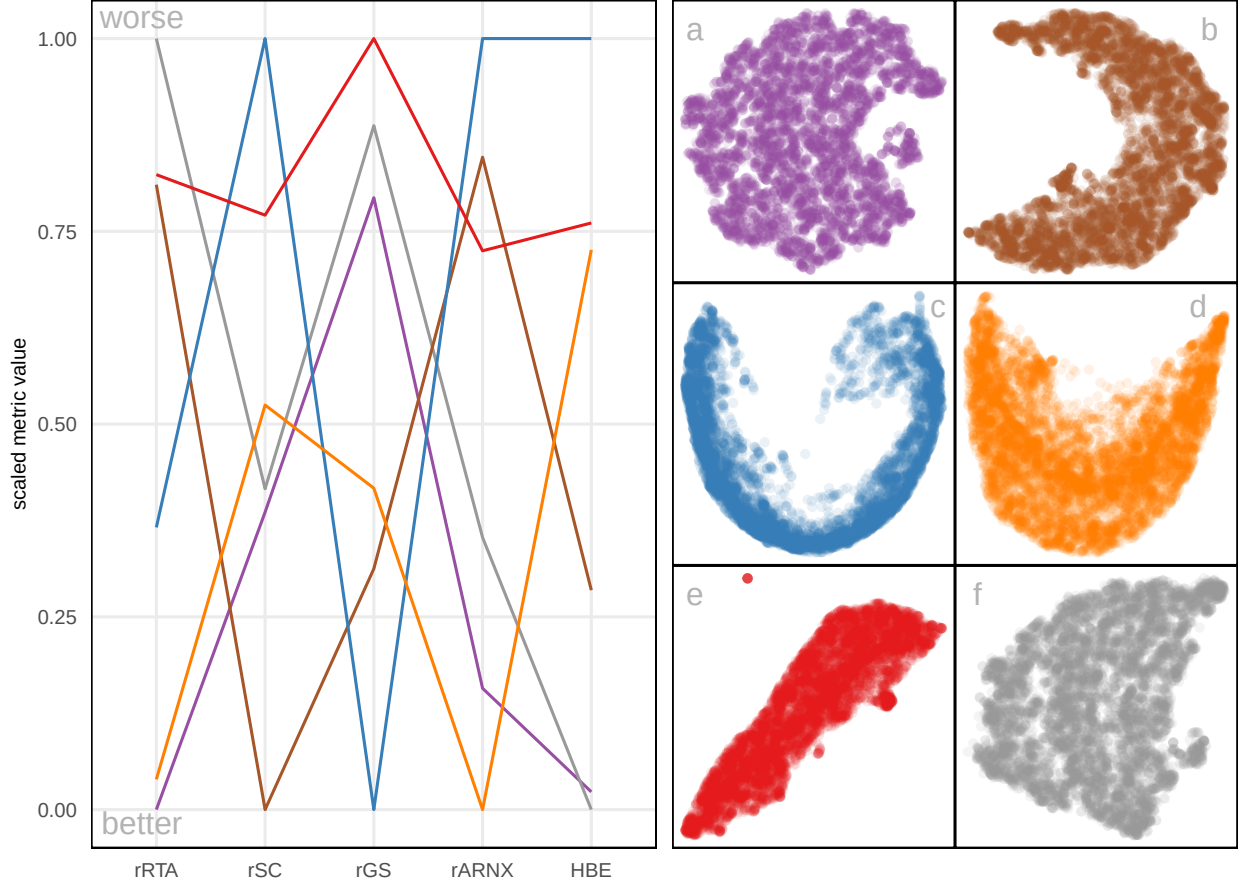


Figure 9: Comparison of scaled evaluation metrics (rARNX, rRTA, rSC, rGS, and HBE using  $a_1 = 0.04$ ) for six NLDR layouts computed on the MNIST digit 1 data using a parallel coordinate plot. Each line represents a layout (a–f), with colors corresponding to the scatterplots shown on the right. The metrics display different ranking patterns, indicating that no single measure fully captures embedding quality. Layout a is identified as the best according to HBE and rRTA, but is considered much less optimal by rARNX, rSC, and rGS. Layout e is considered the worst, or close to the poorest, by all metrics. Layouts a and f show similar patterns of agreement across metrics, except for rRTA, where layout a performs the best and layout f the worst. Layout c is the worst in rARNX, rSC, and HBE.



## References

- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* **4**(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.
- Xia, L., Lee, C. & Li, J. J. (2023), ‘scDEED: A Statistical Method for Detecting Dubious 2D Single-cell Embeddings’, *bioRxiv* . <https://www.biorxiv.org/content/early/2023/04/25/2023.04.21.537839>.